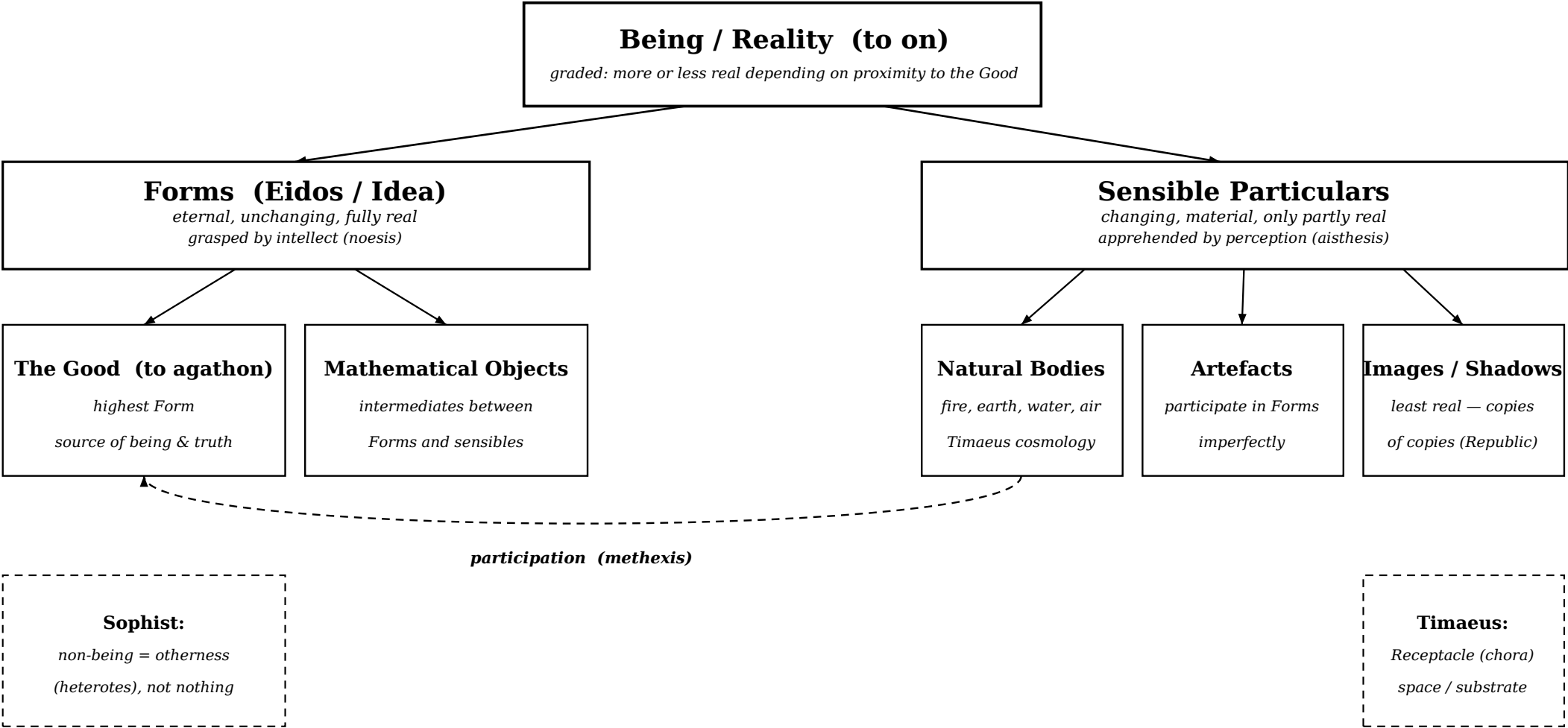




ARISTOTLE

Categories · Metaphysics · De Anima
384-322 BCE

Mind-independent
Endurantist

Being

on · pollachos legetai
said in many ways

cross-cut: Actuality (energeia) vs Potentiality (dynamis) · Metaph. IX

Substance

ousia · exists in its own right

Primary Substance

tode ti · this individual
e.g. Socrates, this horse

Secondary Substance

species & genus
e.g. Human, Animal

form of

Soul (psyche)

De Anima II: form of a
natural body capable of life
not itself a substance

Accident

symbebekos · exists only in a subject · nine categories

Quantity

poson
six feet

Quality

poion
pale

Relation

pros ti
double

Place

pou
Lyceum

Time

pote
yesterday

Position

keisthai
sitting

Having

echein
armed

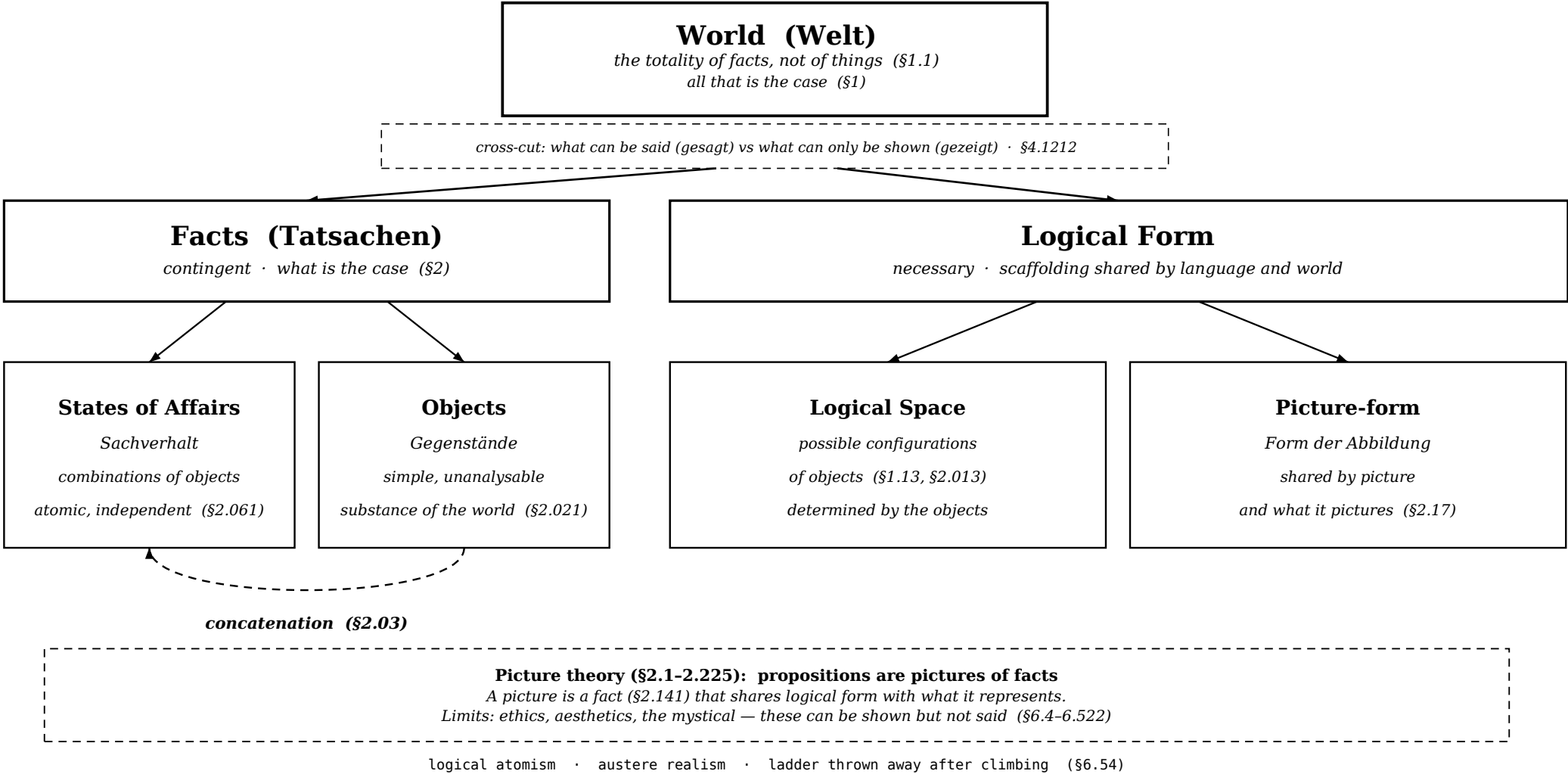
Action

poiein
cutting

Passion

paschein
being cut

Metaphysics: hyle (matter) + morphe (form) compose substance · Form = essence (to ti en einai)



Object (Gegenstand)
anything thinkable · the most general category
every thought has an object, even non-existent ones

cross-cut: Sein (being) vs Sosein (so-being / having properties) — Sosein is independent of Sein

Existieren
existence
concrete, spatiotemporal, causal

Bestehen
subsistence
abstract, ideal, atemporal

Aussersein
beyond-being
has Sosein but no Sein

Physical Things
material objects
events, processes

Mental Acts
Brentano: presentation,
judgment, emotion

**Numbers /
Mathematical**
ideal objects, sets

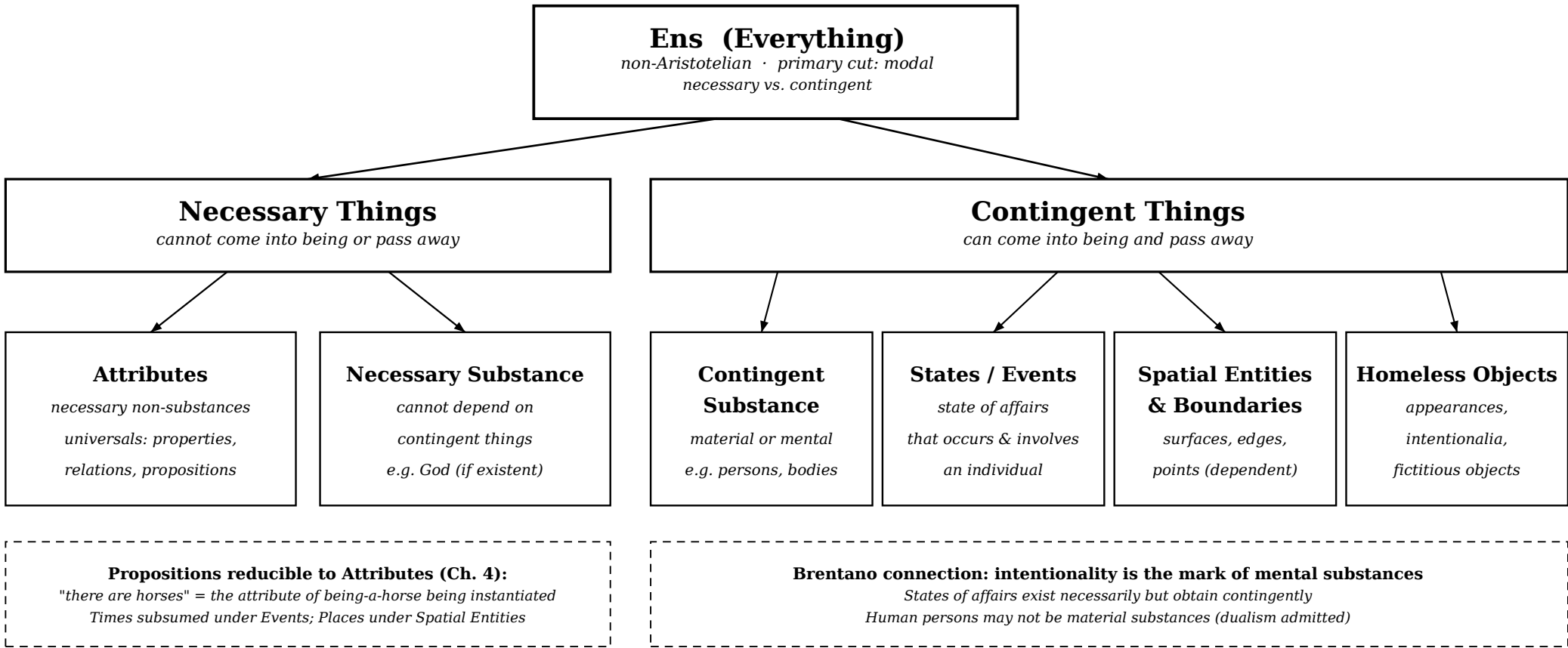
Objectives
Objektive
propositions, states of affairs

**Incomplete /
Impossible**
golden mountain,
round square

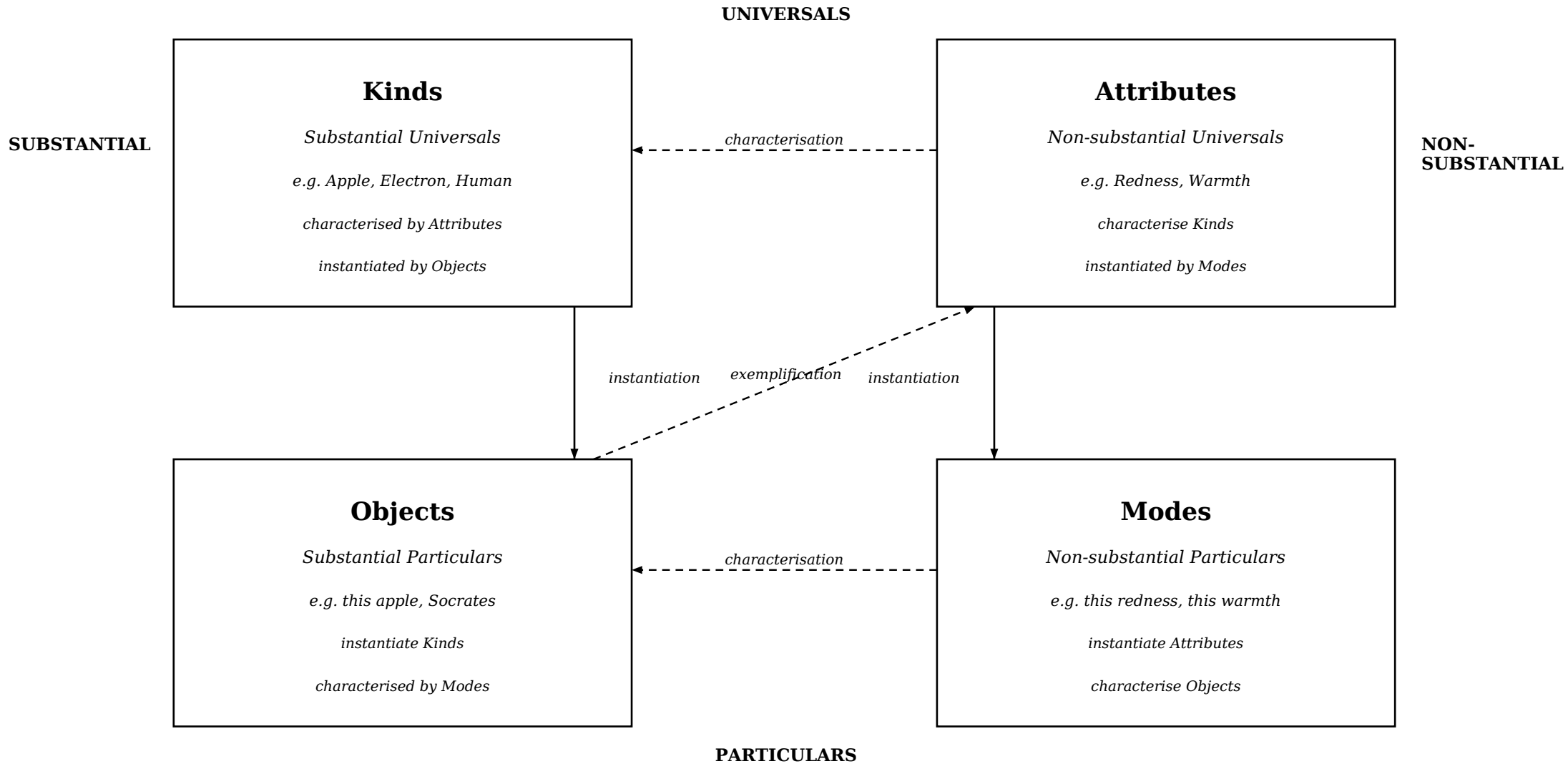
Fictional Objects
Pegasus, Hamlet,
Sherlock Holmes

Russell ("On Denoting", 1905) attacked Aussersein as ontologically extravagant — the founding dispute of analytic philosophy
Downstream: free logic · possible-world semantics · fictional-object theory (Parsons, Zalta)
Already implicit in the deck: Chisholm "Homeless Objects" · EFO worlds (real / fictional / possible)

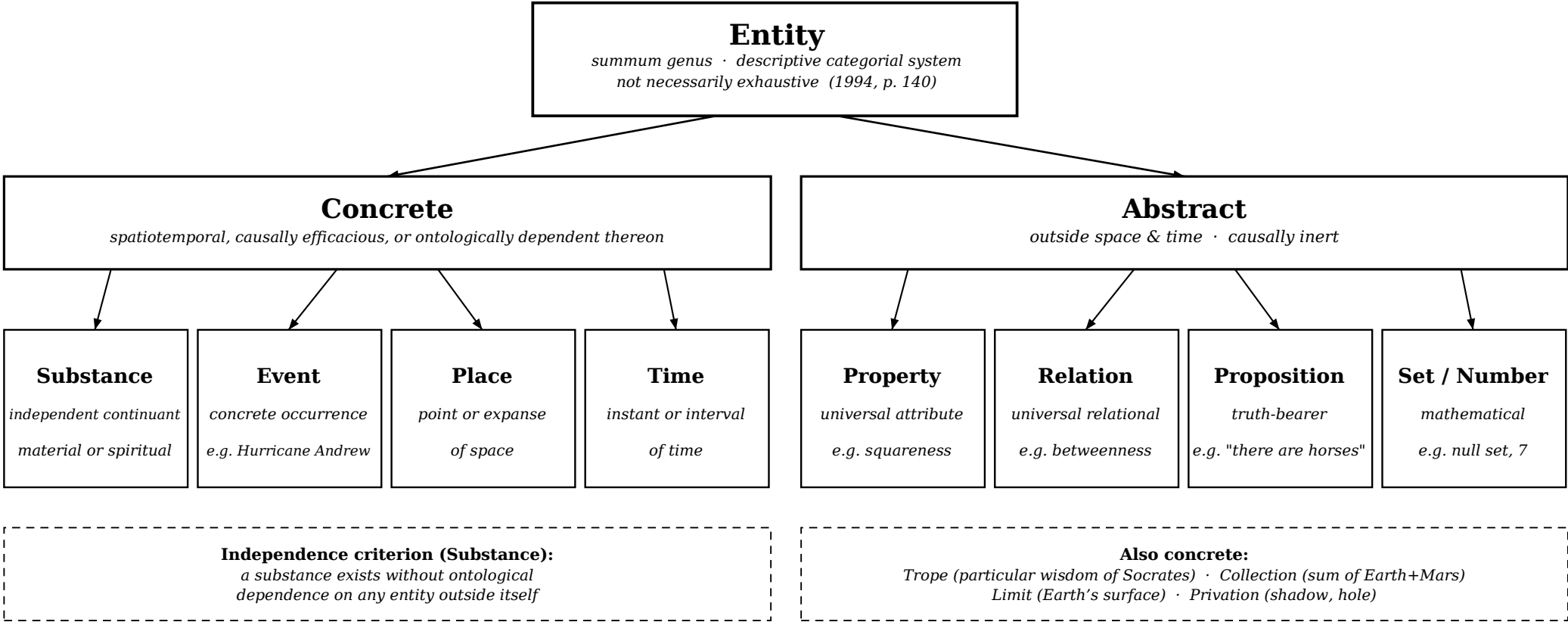
Brentano school (Graz) · maximalist ontology · Sosein independent of Sein · ancestor of EFO thought-objects



The Ontological Square: two cross-cutting distinctions (Substance/Non-substance $\times$ Universal/Particular) generate four categories.
Relations shown: solid arrows = instantiation · dashed arrows = characterisation · diagonal = exemplification



Laws of Nature: a Kind being characterised by an Attribute
e.g. "Electrons have negative charge" = Kind:Electron characterised by Attribute:NegativeCharge



Object in general (Gegenstand überhaupt)

anything that can be the bearer of predicates

cross-cut: Foundation (Fundierung) — dependent vs independent parts · LU III

Formal Ontology

applies to any object whatever · empty of content

Object

*Gegenstand
individual*

State of Affairs

Sachverhalt

Whole / Part

*mereology
LU III*

Relation

*Beziehung
unity, plurality*

Material (Regional) Ontologies

a priori structures of specific domains of being

Nature

*physical things
Ideas II §1*

Animate Nature

*psyche, body,
lived experience*

Spirit / Culture

*Geist · social,
historical world*

Pure Logic

*ideal meanings,
Bedeutungen*

founded on (fundiert in)

coined "formal ontology" · ancestor of BFO, DOLCE, GFO mereology & dependence

BFO

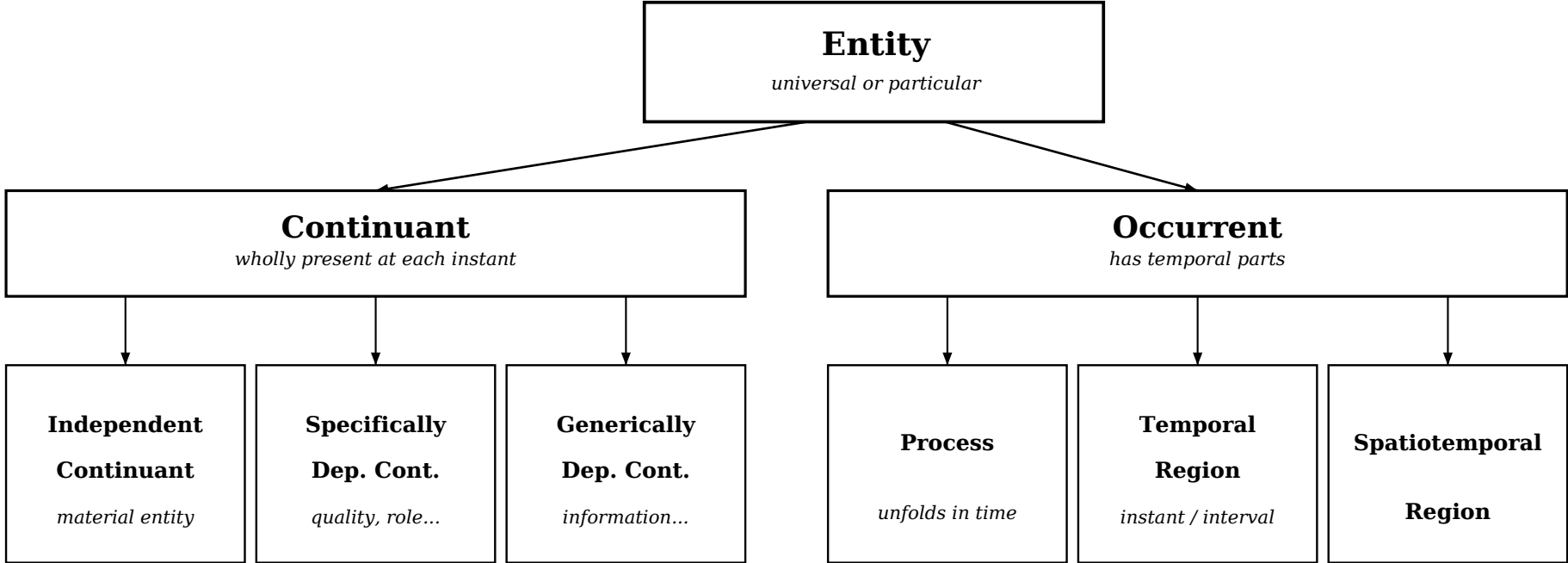
Basic Formal Ontology
Arp, Smith, Spear · 2015

Mind-independent

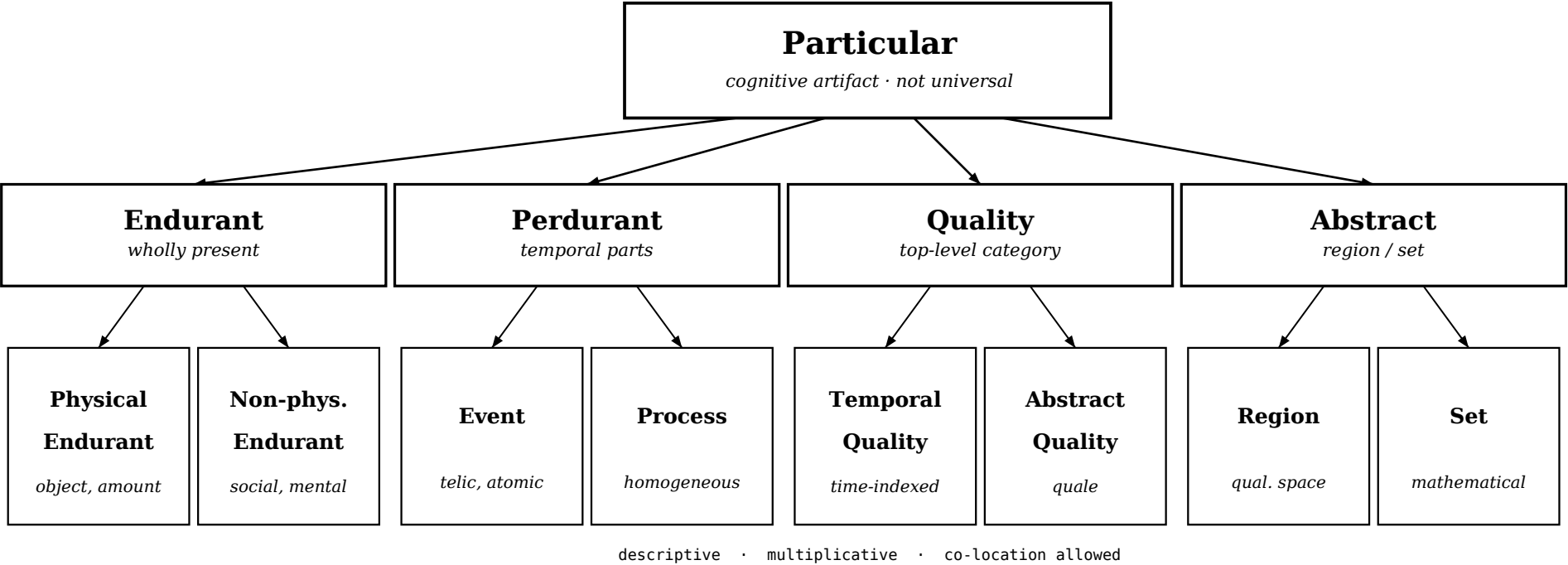
Aristotelian

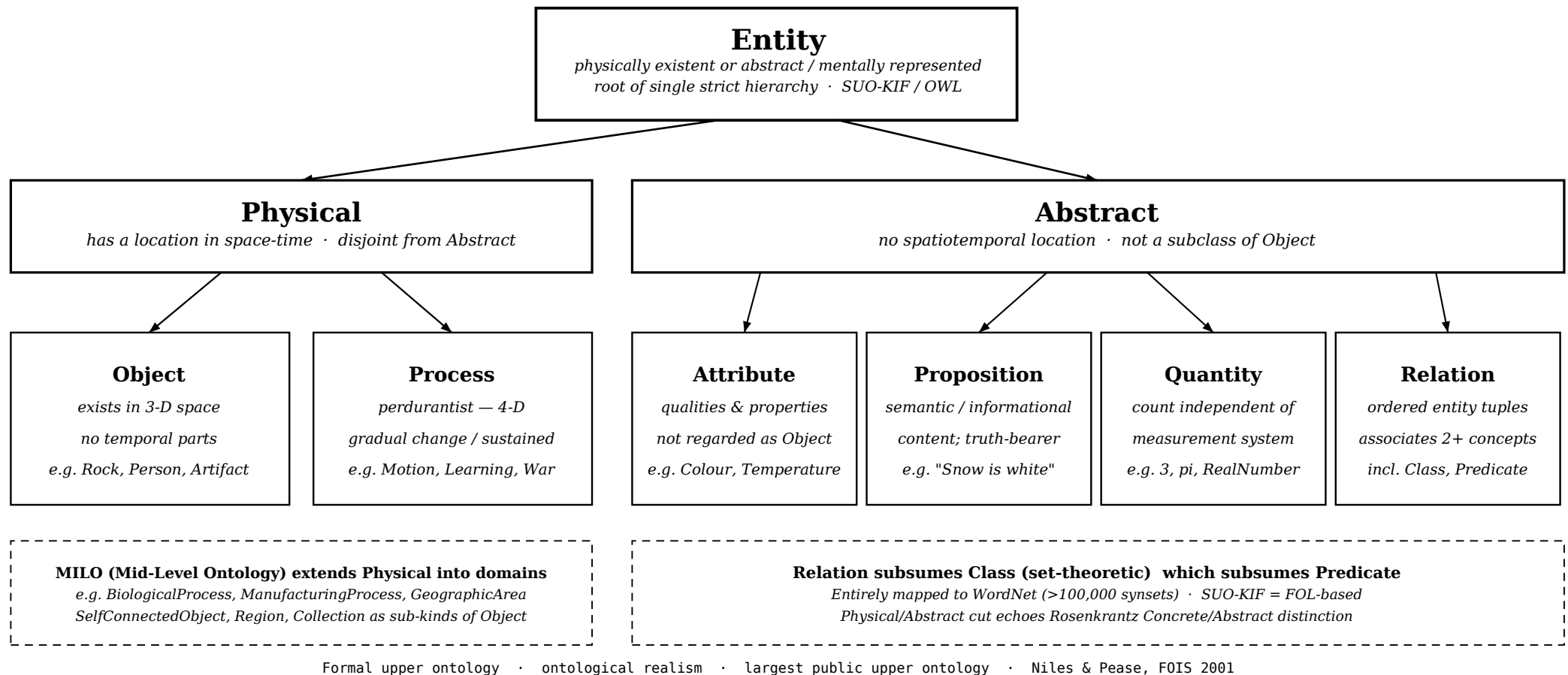
Formalized

Endurantist + Perdurantist



ontological realism · universals in re





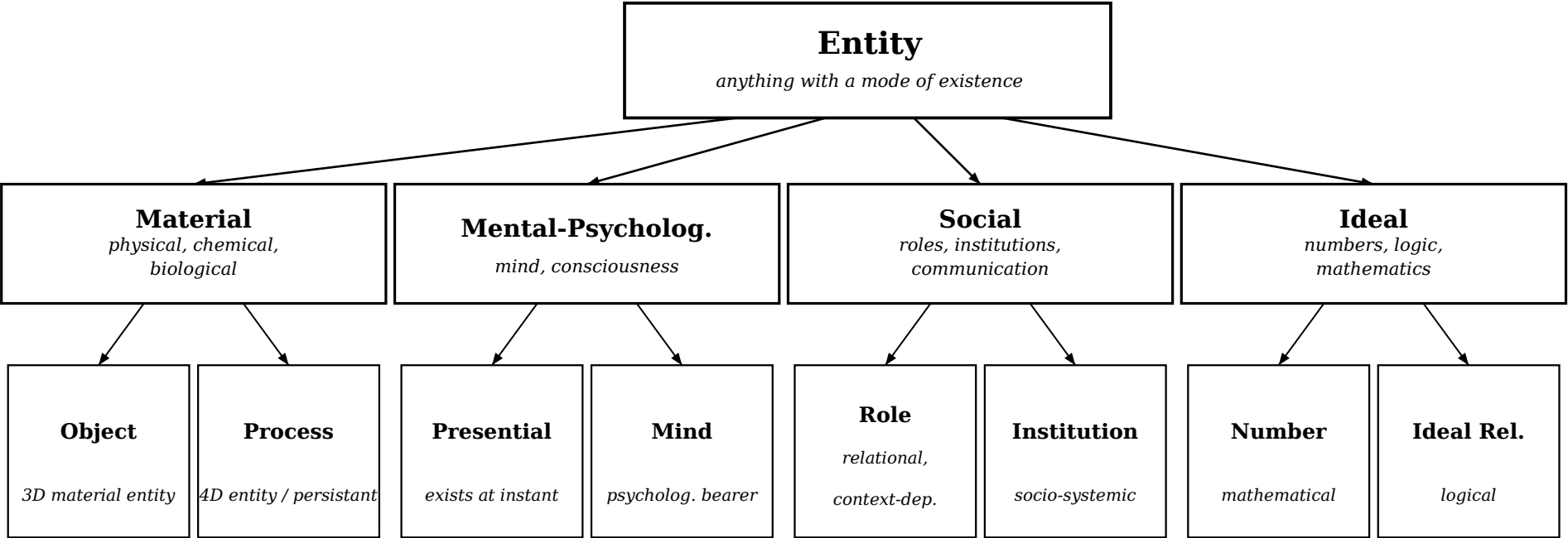
GFO

General Formal Ontology
Herre et al. · Leipzig · 1999-

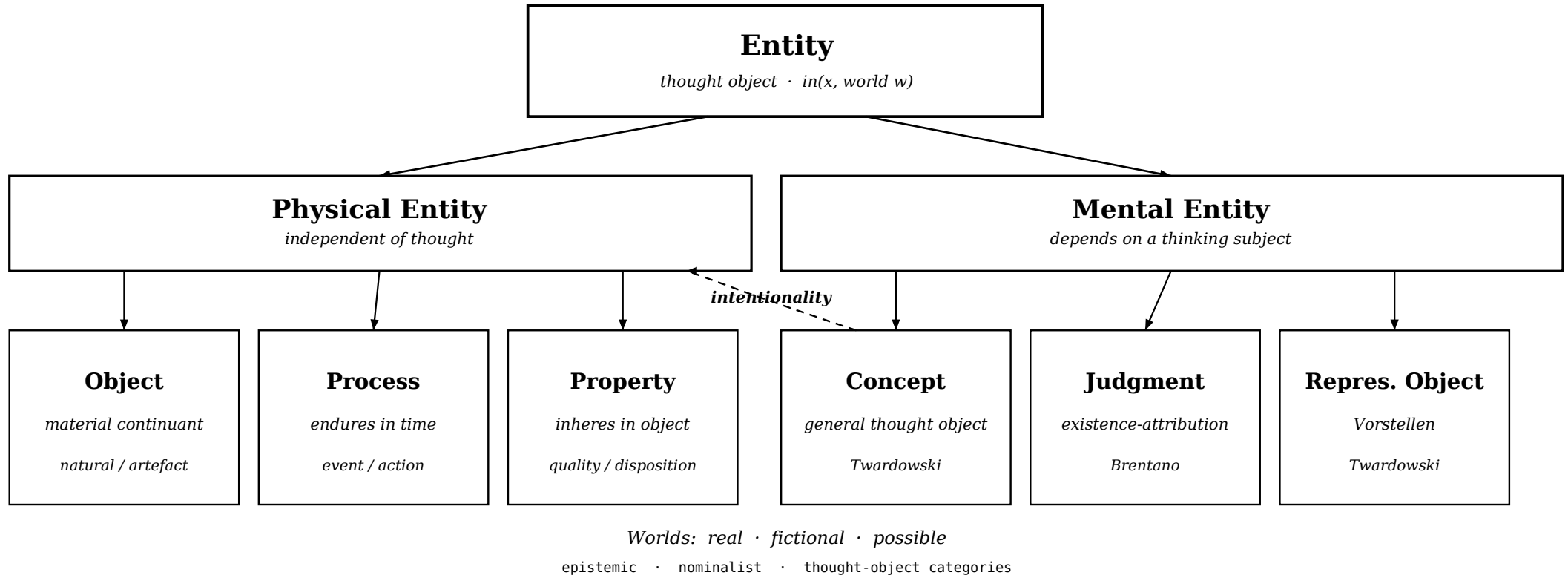
Mind-inclusive

Formalized

Endurantist + Perdurantist



cross-cutting: Individual vs. Category (GFO term "Categor" = universal / concept / symbol structure)
4 ontological strata · 3-layer meta-ontology · integrates objects + processes



RADICAL CONSTRUCTIVISM

A Way of Knowing and Learning · Ernst von Glasersfeld
Falmer Press · 1995 (essays from the 1970s onward)

Non-Mind-independent

Non-Mind-inclusive

Non-Representational

Cognition is adaptive, not representational.

Knowledge is viable (it works), not true (it mirrors reality).

There is no categorial scheme — categories are constructions, contingent on the cognizing subject.

Traditional ontology assumes...

the assumption shared by all twelve preceding pages

Categories track reality

kinds carve the world at its joints

Truth = correspondence

a proposition is true iff it matches reality

Knowledge = representation

mind mirrors what is the case

Cognition is finding

the world is given; we discover its structure

Reality is independently accessible

we can know how things are in themselves

Radical Constructivism claims...

Glasersfeld's counter-position

Categories are constructed

by the cognizing subject, in coordinating experience

Viability replaces truth

a construct survives if it works — like a key fitting a lock

Cognition is adaptive

mind builds models that fit experience, not reality

Cognition is making

verum ipsum factum (Vico): we know what we make

Reality is inaccessible as such

whatever is "out there" cannot be known apart from our constructions

Lineage: Vico (1710) → Kant (constituting categories) → Piaget (genetic epistemology) → von Foerster, Maturana (cybernetics) → Glasersfeld

Downstream: constructivist pedagogy (Piaget, Papert) · second-order cybernetics · autopoietic theory (Maturana & Varela) · enactivism

NOT: social constructionism (Berger & Luckmann) · postmodern relativism — RC is an epistemology, not a sociology of knowledge

meta-position · no categorial tree · questions the very enterprise of the preceding 12 pages